Integrals Practice Answer Key and Solutions

Grade 12 | Free

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This free set moves from antiderivatives to definite integrals and reasoning. Attempt questions before reading the companion solutions. All angles are in radians; ln is the natural logarithm. Give exact answers. Indefinite integrals include C and are considered on intervals where their integrands are defined. $\lfloor u \rfloor$ denotes the greatest integer not exceeding u.

Answer key

1. $7x^4/4 - 2x^2 + 9x + C$

2. $\frac{10}{3}x^{3/2} + 3/x + C$

3. $F(x) = 2x^3 - 5x + 1$

4. $\frac{4}{3}\sin 3x + \frac{5}{2}\cos 2x + C$

5. $\frac{1}{2}e^{4x} + 3^x/\ln 3 + C$

6. $\ln(3x^2 + 8) + C$

7. $(2x^2 + 5)^5/20 + C$

8. $\ln(4 + \sin x) + C$

9. $\frac{1}{6}\tan^{-1}(2x/3) + C$

10. $\frac{1}{3}\sin^{-1}(3x/5) + C$

11. $\ln(x^2 + 7x + 15) + C$

12. $\frac{1}{2}\tan^{-1}((x+3)/2) + C$

13. $\frac{1}{3}\ln\left|\frac{x+2}{x+5}\right| + C$

14. $2\ln|x+2| + \ln|x+3| + C$

15. $x - \tan^{-1}(x/2) + C$

16. $e^{3x}(3x-1)/9 + C$

17. $-x\cos 4x/4 + \sin 4x/16 + C$

18. $x\ln(2x) - x + C$

19. $\frac{x^3}{3}\ln x - \frac{x^3}{9} + C$

20. $x/2 - \sin 6x/12 + C$

21. $\sin^4 x/4 + C$

22. $\tan^3 x/3 + C$

23. $2\sqrt{x} - 4\ln(\sqrt{x} + 2) + C$

24. $-1/(3 + \ln x) + C$

25. $2\sqrt{x^2 + x + 6} + C$

26. 15

27. $\frac{1}{2}\ln(9/5)$

28. $1/3$

29. $1/4$

30. 0

31. $144/5$

32. $5/2$

33. $13/2$

34. $13/2$

35. 5

36. $\ln 2 - 1/2$

37. $(3e^4 + 1)/4$

38. $\frac{1}{2}\ln(3/2)$

39. $2x(1 + x^8)$

40. $\frac{3}{1+9x^2} - \frac{1}{1+x^2}$

41. $\ln(4/3)$

42. $1 - \pi/4$

43. $9/5$

44. $9/4$, attained at $a = 1/2$

45. $(1 - \ln 2)/2$

46. $\pi/12$

47. $\frac{6}{(n+1)(n+2)(n+3)(n+4)}$

48. $f(x) = 3e^{x^2}$; integral $= \frac{3}{2}(e - 1)$

49. The stated strict bounds hold.

50. $x = 1$ is a local maximum; $x = 3$ is a local minimum.

Question 1
Answer: $7x^4/4 - 2x^2 + 9x + C$
Integrate each term separately. The power rule raises the exponent by one and divides by that new exponent. Thus $7(x^4/4) - 4(x^2/2) + 9x + C$ is the required family. Differentiation gives the original integrand.

Question 2
Answer: $\frac{10}{3}x^{3/2} + 3/x + C$
Write the integrand as $5x^{1/2} - 3x^{-2}$. Its integral is $5x^{3/2}/(3/2) - 3x^{-1}/(-1) + C$.

Question 3
Answer: $F(x) = 2x^3 - 5x + 1$
First $F(x) = 2x^3 - 5x + C$. At $x = 2$, $7 = 16 - 10 + C$, so $C = 1$.

Question 4
Answer: $\frac{4}{3}\sin 3x + \frac{5}{2}\cos 2x + C$
The derivative of $\sin 3x$ is $3\cos 3x$, and that of $\cos 2x$ is $-2\sin 2x$. Divide by these inner derivatives.

Question 5
Answer: $\frac{1}{2}e^{4x} + 3^x/\ln 3 + C$
Use $\int e^{ax}dx = e^{ax}/a$ and $\int b^x dx = b^x/\ln b$. Both denominators are necessary.

Question 6
Answer: $\ln(3x^2 + 8) + C$
Put $u = 3x^2 + 8$; then $du = 6x\,dx$. The integral becomes $\int du/u = \ln u + C$. Here $u > 0$.

Question 7
Answer: $(2x^2 + 5)^5/20 + C$
With $u = 2x^2 + 5$, $du = 4x\,dx$. Hence the integral is $\frac{1}{4}\int u^4 du = u^5/20 + C$.

Question 8
Answer: $\ln(4 + \sin x) + C$
Choose $u = 4 + \sin x$ so that $du = \cos x\,dx$. Since $u \geq 3$, the logarithm is defined without an absolute-value sign.

Question 9
Answer: $\frac{1}{6}\tan^{-1}(2x/3) + C$
Put $u = 2x/3$, so $dx = 3du/2$ and the denominator is $9(1 + u^2)$. This gives $\frac{1}{6}\int du/(1 + u^2)$.

Question 10
Answer: $\frac{1}{3}\sin^{-1}(3x/5) + C$
Let $u = 3x/5$. Then $dx = 5du/3$ and the root is $5\sqrt{1 - u^2}$. Integrate the inverse-sine form.

Question 11
Answer: $\ln(x^2 + 7x + 15) + C$
The numerator is the derivative of the denominator. The denominator is $(x + 7/2)^2 + 11/4 > 0$, so direct logarithmic substitution applies.

Question 12
Answer: $\frac{1}{2}\tan^{-1}((x + 3)/2) + C$
Complete the square: $x^2 + 6x + 13 = (x + 3)^2 + 4$. Set $u = x + 3$ and use the arctangent formula with $a = 2$.

Question 13
Answer: $\frac{1}{3}\ln\left|\frac{x+2}{x+5}\right| + C$
Write $1/[(x+2)(x+5)] = \frac{1}{3}[1/(x+2) - 1/(x+5)]$. Integrate on an interval avoiding $-2, -5$.

Question 14
Answer: $2\ln|x+2| + \ln|x+3| + C$
Factor the denominator. Solve $3x+8 = A(x+3) + B(x+2)$ to obtain $A = 2, B = 1$. Integrate the two simple fractions.

Question 15
Answer: $x - \tan^{-1}(x/2) + C$
Division gives $1 - 2/(x^2+4)$. Since $\int dx/(x^2+4) = \frac{1}{2}\tan^{-1}(x/2)$, the result follows.

Question 16
Answer: $e^{3x}(3x-1)/9 + C$
Choose $u = x$, $dv = e^{3x}dx$, giving $du = dx$, $v = e^{3x}/3$. Then $uv - \int v du = xe^{3x}/3 - e^{3x}/9 + C$.

Question 17
Answer: $-x\cos 4x/4 + \sin 4x/16 + C$
Integration by parts with $u = x$ and $v = -\cos 4x/4$ gives $-x\cos 4x/4 + \frac{1}{4}\int \cos 4x\, dx$.

Question 18
Answer: $x\ln(2x) - x + C$
Use $u = \ln(2x)$, $dv = dx$. As $du = dx/x$, the remaining integral is $\int 1\, dx = x$.

Question 19
Answer: $\frac{x^3}{3}\ln x - \frac{x^3}{9} + C$
Take $u = \ln x$, $dv = x^2 dx$. Subtract $\frac{1}{3}\int x^2 dx$ from $x^3\ln x/3$.

Question 20
Answer: $x/2 - \sin 6x/12 + C$
Use $\sin^2(3x) = (1 - \cos 6x)/2$. Integrate the constant term and cosine term separately.

Question 21
Answer: $\sin^4 x/4 + C$
With $u = \sin x$, $du = \cos x\, dx$, the integral is $\int u^3 du = u^4/4 + C$.

Question 22
Answer: $\tan^3 x/3 + C$
Put $u = \tan x$; $du = \sec^2 x\, dx$. Integrate u^2.

Question 23
Answer: $2\sqrt{x} - 4\ln(\sqrt{x}+2) + C$
Put $t = \sqrt{x}$, $dx = 2t\, dt$. Divide: $2t/(t+2) = 2 - 4/(t+2)$. Integrate and replace t.

Question 24
Answer: $-1/(3+\ln x) + C$
Set $u = 3 + \ln x$, so $du = dx/x$. Then $\int u^{-2} du = -u^{-1} + C$.

Question 25
Answer: $2\sqrt{x^2+x+6} + C$
Set $u = x^2 + x + 6$, $du = (2x+1)dx$. Integrating $u^{-1/2}$ gives $2\sqrt{u}$.

Question 26
Answer: 15
An antiderivative is $x^3 + 2x$. Subtract its value -3 at -1 from its value 12 at 2.

Question 27
Answer: $\frac{1}{2}\ln(9/5)$
Put $u = x^2 + 5$. The limits become $5, 9$ and the integral becomes $\frac{1}{2}\int_5^9 du/u$.

Question 28
Answer: $1/3$
An antiderivative is $-\cos 3x/3$. Its endpoint difference is $0 - (-1/3)$.

Question 29
Answer: $1/4$
Use $-1/(x+1)$. The difference is $-1/4 + 1/2 = 1/4$.

Question 30
Answer: 0
The numerator is odd and the denominator is even and positive. Therefore the integrand is continuous and odd, so its integral on the symmetric interval vanishes.

Question 31
Answer: $144/5$
The integrand is even. Twice the integral from 0 to 2 is $2[x^5/5 + x^3]_0^2 = 2(32/5 + 8) = 144/5$.

Question 32
Answer: $5/2$
Call the integral I and replace x by $5 - x$. The reflected numerator is $(6 - x)^2$. Adding the two forms gives $2I = \int_0^5 1\,dx = 5$.

Question 33
Answer: $13/2$
The sign changes at $x = 1/2$. Hence $I = \int_0^{1/2}(1 - 2x)dx + \int_{1/2}^3(2x - 1)dx = 1/4 + 25/4 = 13/2$.

Question 34
Answer: $13/2$
Split at 1. The first part is $[x^2/2 + 2x]_0^1 = 5/2$. The second is $[4x - x^2/2]_1^3 = 4$. Add them.

Question 35
Answer: 5
On the five intervals $[k/2, (k+1)/2)$ the value is k, for $k = 0, 1, 2, 3, 4$. Sum the rectangle areas: $(0 + 1 + 2 + 3 + 4)/2 = 5$. Endpoint values do not alter the integral.

Question 36
Answer: $\ln 2 - 1/2$
Put $u = 1 + x^2$. Then $I = \frac{1}{2}\int_1^2 \ln u\,du = \frac{1}{2}[u\ln u - u]_1^2 = \ln 2 - 1/2$.

Question 37
Answer: $(3e^4 + 1)/4$
Parts gives $e^{2x}(2x - 1)/4$. At the endpoints the values are $3e^4/4$ and $-1/4$. Subtract.

Question 38
Answer: $\frac{1}{2}\ln(3/2)$
Let $u = 2 + \sin^2 x$, $du = 2\sin x \cos x\, dx$. The transformed bounds are $2, 3$.

Question 39
Answer: $2x(1 + x^8)$
The fundamental theorem gives the integrand at the upper limit, multiplied by the derivative of that limit. Thus $G'(x) = (1 + (x^2)^4)(2x)$.

Question 40
Answer: $\frac{3}{1+9x^2} - \frac{1}{1+x^2}$
The upper endpoint contributes $3/(1 + 9x^2)$. Subtract the lower endpoint contribution $1/(1 + x^2)$.

Question 41
Answer: $\ln(4/3)$
This is a right-endpoint Riemann sum on $[0, 1]$ for $f(x) = 1/(3 + x)$. The limit is $\int_0^1 dx/(3 + x) = [\ln(3 + x)]_0^1$.

Question 42
Answer: $1 - \pi/4$
Each term equals $\frac{1}{n}\frac{(r/n)^2}{1+(r/n)^2}$. Integrate $x^2/(1 + x^2) = 1 - 1/(1 + x^2)$ from 0 to 1.

Question 43
Answer: $9/5$
Expand and integrate: $\int_0^3 (x^2 + x)dx = 27/2$ and $\int_0^3 (x + 1)dx = 15/2$. Thus $27/2 - c(15/2) = 0$.

Question 44
Answer: $9/4$, attained at $a = 1/2$
Integration gives $J(a) = 3a^2 - 3a + 3 = 3(a - 1/2)^2 + 9/4$. A square is nonnegative, establishing both the minimum and its unique location.

Question 45
Answer: $(1 - \ln 2)/2$
Divide $x^3/(1 + x^2) = x - x/(1 + x^2)$. The first integral is $1/2$ and the second is $\frac{1}{2}\ln 2$.

Question 46
Answer: $\pi/12$
Use $t = \tan x$ for $x < \pi/2$ and take the limit as $t \to \infty$. The integral is $\int_0^\infty dt/(9 + 4t^2) = [\frac{1}{6}\tan^{-1}(2t/3)]_0^\infty = \pi/12$. The original integrand is continuous at the endpoint.

Question 47
Answer: $\frac{6}{(n+1)(n+2)(n+3)(n+4)}$
Integrate by parts with $dv = x^n dx$ and $u = (1 - x)^3$. The endpoint product vanishes, giving $I_n = \frac{3}{n+1}\int_0^1 x^{n+1}(1 - x)^2 dx$. Repeat twice, then use $\int_0^1 x^{n+3}dx = 1/(n + 4)$.

Question 48
Answer: $f(x) = 3e^{x^2}$; integral $= \frac{3}{2}(e - 1)$
Differentiate to get $f' = 2xf$ with $f(0) = 3$. Equivalently $(e^{-x^2} f(x))' = 0$, so $f(x) = 3e^{x^2}$. Substitute $u = x^2$ in the requested integral.

Question 49
Answer: The stated strict bounds hold.
For $0 < x < 1$, $5 < 5 + x^4 < 6$, so $1/6 < 1/(5 + x^4) < 1/5$. Integrate over the unit interval. The inequalities are strict on an interval of positive length, so equality is impossible.

Question 50
Answer: $x = 1$ is a local maximum; $x = 3$ is a local minimum.
$F'(x) = (x - 1)(x - 3)$ is positive for $x < 1$, negative for $1 < x < 3$, and positive for $x > 3$. The two sign changes give the classification. Integration gives $F = x^3/3 - 2x^2 + 3x$, so the stationary values are $4/3$ and 0.